

MARKOV CHAIN MONTE CARLO

From Metropolis - Hastings to Bayesian regression

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MCMC emerged from an impossible integration problem



1 ■ HISTORICAL BACKGROUND

- ▶ **Los Alamos, 1949 - 1952.** Early computers - SEAC, JOHNNIAC and MANIAC - enabled Monte Carlo studies of complex many-body systems.
- ▶ **Challenge.** Equilibrium properties required intractable integration over vast configuration spaces.
- ▶ **Insight.** Generate representative configurations with probabilistic acceptance rules - rather than explore every state.

Metropolis acceptance rule

$\Delta E < 0 \Rightarrow$ accept,

$\Delta E > 0 \Rightarrow$ accept w.p. $e^{-\Delta E/T}$.

Key outcome

The chain converges to the equilibrium distribution:

$$P(x) \propto e^{-E(x)/T}$$

Source: Metropolis et al. (1953).

Hastings turned a physics algorithm into a general method

1 ■ FROM PHYSICS TO STATISTICS

Metropolis (1953) sampled equilibrium states; Hastings (1970) extended the idea to arbitrary inferential targets.

Factorisation

Required conditional densities may be unavailable.

Importance sampling

Poor proposals produce unstable weights.

Rejection sampling

Wide proposal envelopes cause low acceptance.

The gap Hastings closed

Target the required distribution using density *ratios*; the normalising constant cancels.

Source: Hastings (1970), p. 98.

MCMC replaces an integral with a convergent average

2 ■ MATHEMATICAL DEVELOPMENT

Monte Carlo estimator

$$\theta = E_{\pi}[h(X)] = \int h(x)\pi(x) dx$$

Given draws $X_1, \dots, X_N$ from π ,

$$\hat{\theta} = \frac{1}{N} \sum_{i=1}^N h(X_i) \longrightarrow E_{\pi}[h(X)].$$

Although dependent, MCMC averages converge by the ergodic theorem.

Source: Gelman et al. (2013).

Bayesian posterior target

$$p(\theta | y) = \frac{p(y | \theta)p(\theta)}{\int p(y | \theta)p(\theta) d\theta}.$$

The denominator is constant in θ , so

$$\pi(\theta | y) \propto L(\theta | y)p(\theta).$$

Posterior ratios cancel the unknown normalising constant.

Detailed balance makes the target distribution stationary

2 ■ MATHEMATICAL DEVELOPMENT

Balance condition

$$\pi(x)K(x, x') = \pi(x')K(x', x).$$

Metropolis–Hastings enforces this with

$$\alpha(x, x') = \min \left\{ 1, \frac{\pi(x')q(x | x')}{\pi(x)q(x' | x)} \right\}.$$

With a symmetric proposal, q cancels, recovering Metropolis.

Source: Hastings (1970).

Sampling loop

1. Initialise $x^{(0)}$.
2. Propose $x^* \sim q(x^* | x^{(t)})$.
3. Calculate α .
4. Draw $u \sim U(0, 1)$.
5. If $u \leq \alpha$, accept x^* ; otherwise retain $x^{(t)}$.
6. Repeat; discard warm-up; summarise retained draws.

Valid inference rests on seven theoretical conditions

2 ■ MATHEMATICAL DEVELOPMENT

Markov property

Next state depends only on the current state.

Irreducibility

Every positive-probability region is reachable.

Aperiodicity

No deterministic cycling.

Detailed balance

Equilibrium probability flow is balanced between states.

Stationarity

Transition kernel leaves the target unchanged.

Ergodicity

Long-run averages converge to target expectations.

Proposal support

Proposal reaches every relevant target region.

The application uses 768 diabetes observations

3 ■ DATA DESCRIPTION

- ▶ **Source:** NIDDK; UCI Repository; R dataset PimaIndiansDiabetes2.
- ▶ **Population:** 768 Pima Indian women, aged 21+.
- ▶ **Response:** oral-test plasma glucose (mg/dL).
- ▶ **Final predictors:** pregnancies, blood pressure, triceps, insulin, pedigree and age.

Source: UCI repository.

9 variables

Eight clinical measures + one diabetes-status indicator.

Modelling decision

BMI was imputed but excluded from the final regression; diabetes status was descriptive.

Multiple imputation preserves information in incomplete records

3 ■ DATA DESCRIPTION

Extent of missingness

Variable	Missing
Insulin	49.0%
Triceps	30.0%
Blood pressure	4.6%
BMI	1.4%
Glucose	0.7%

MICE predictive mean matching under MAR:
 $m = 20$ completed datasets.

Strongest reported correlations

Triceps–BMI: $r = 0.623$

Glucose–insulin: $r = 0.604$

Pregnancies–age: $r = 0.544$

All predictor VIF values were below 5.

Regression diagnostics

Residuals had mild tail departure and were approximately normal. Raw glucose was heteroscedastic ($p < 0.05$); log-glucose was not ($p > 0.05$).

Three estimators were fitted to the same imputed data

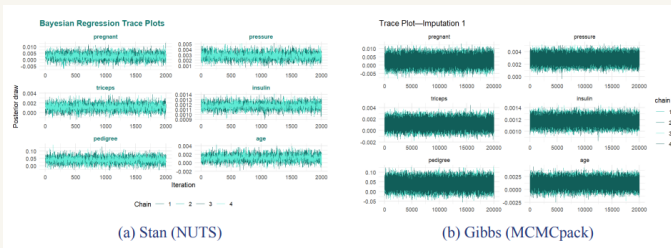
4 ■ IMPLEMENTATION

- ▶ **Pre-processing:** implausible zeros set to missing; MICE-PMM produced 20 datasets (seed 123; up to 50 iterations).
- ▶ **Classical regression:** pooled six-predictor OLS for log-glucose.
- ▶ **Gibbs sampling:** conjugate Gaussian MCMCregress; four chains and 80,000 retained draws per imputation.
- ▶ **Stan / NUTS:** `stan_glm`; four chains, 4,000 iterations and 2,000 warm-up.

Diagnostics: trace plots ■ $\hat{R}$ ■ effective sample size ■ autocorrelation ■ posterior predictive checks

Independent chains converged to the same posterior region

4 ■ IMPLEMENTATION



$$\hat{R} \approx 1.00$$

Near-one values show strong between-chain agreement.

Large effective samples

Gibbs: 75,805–83,107.
Stan: 6,396–8,874 per predictor.

Trace plot Interpretation

Stable, overlapping traces with no drift support convergence and mixing.

The chains mix efficiently and reproduce the observed outcome

4 ■ IMPLEMENTATION

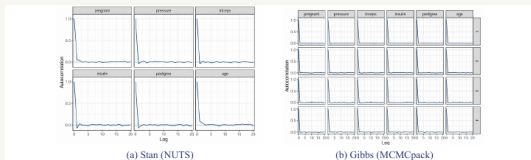
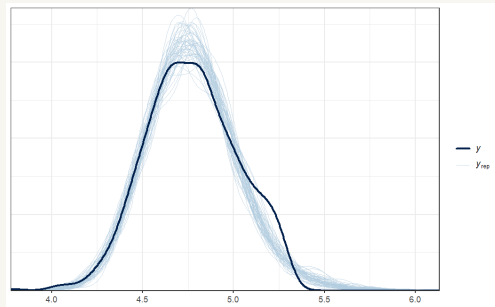


Figure 1: Autocorrelation comparison for the Stan/NUTS (a) and Gibbs (b) sampling models, showing rapid decay to zero within one to two lags for all six predictors consistent with the ergodicity assumption underlying valid MCMC inference

Rapid autocorrelation decay across all six coefficients indicates efficient exploration.



Replicated and observed glucose distributions broadly overlap, supporting model adequacy.

What holds up, and what does not

6 ■ CRITICAL EVALUATION

Theoretical strengths

- ▶ **Rigorous foundation.** Detailed balance makes $\pi(x)$ stationary.
- ▶ **Works with unnormalised densities.** No intractable normalising constant is needed.
- ▶ **Built-in error assessment.** Block methods estimate variance from dependent draws.
- ▶ **Conceptually simple.** Propose, accept/reject, repeat.

Methodological weaknesses

- ▶ **Arbitrary proposal tuning.** No universal proposal step-size rule.
- ▶ **Diagnostics have blind spots.** $\hat{R}$ needs multiple chains; ESS may miss poor tails; trace plots are partly subjective.
- ▶ **Ad hoc burn-in.** Too little retains transients; too much wastes computation.
- ▶ **Correlated samples.** Less information than equal-sized independent samples.

Where Metropolis–Hastings struggles

6 ■ CRITICAL EVALUATION

- ▶ **The curse of dimensionality.** In high dimensions, random-walk steps must shrink to preserve acceptance, slowing exploration.
- ▶ **Autocorrelation inflates variance.** For an AR(1)-like chain with $\rho = 0.9$, ESS may fall to roughly $N/19$.
- ▶ **Multi-modal targets.** Small proposals may trap chains; large proposals may be rejected.

The method can fail when

- ▶ Improper priors yield improper posteriors.
- ▶ Chains are non-ergodic or periodic.
- ▶ The target has infinite variance.
- ▶ Modes are separated by near-zero density.

The methods agree on effects but differ in cost and flexibility

5 ■ COMPARISON WITH MODERN METHODS

Criterion	Classical OLS	Gibbs	Stan / NUTS
Output	Estimate, SE and confidence interval	Full posterior	Full posterior
Strength	Fast, familiar baseline	Efficient under conjugacy	Flexible with rich diagnostics
Limitation	No posterior distribution	Conjugacy-dependent	Greater computational cost
Runtime	Fastest	22.52 seconds	313.20 seconds

Pedigree coefficient: approximately 0.041 across all three estimators.

Gibbs: approximately 13.9× faster than Stan in this conjugate Gaussian model.

Modern samplers replace random walks with informed movement

LIMITATIONS ■ LATER DEVELOPMENTS

Original limitation

Proposal tuning and random walks scale poorly in complex posterior geometries.

What came next

HMC uses gradients; NUTS adapts path length; INLA accelerates selected latent Gaussian models.

What methodological improvements should reviewers expect

- ▶ Dispersed initial values; multiple chains
- ▶ Bulk/tail ESS with $\hat{R}$
- ▶ HMC/NUTS divergence and energy checks
- ▶ Prior sensitivity analysis
- ▶ Predictive checks and out-of-sample validation
- ▶ Reproducible code, processing and seeds

METHOD CHOICE

GIBBS

Conjugate Gaussian models when speed matters.

STAN / NUTS

Hierarchical, non-conjugate and generalised models.

CLASSICAL OLS

Rapid, familiar comparison benchmark.

Trust the posterior only after you trust the chain.

Metropolis - Hastings reached difficult distributions. Gibbs made conjugate Bayesian models practical. Modern diagnostics make conclusions defensible.

GROUP 4 ■ QUESTIONS & DISCUSSION